CSE4107, Spring 2021
Class Test 2
Course Outline

1. Describe the basic concepts of solving AI problems as search problems.
2. What do you mean by informed or heuristic search? Cite examples of heuristic functions.
3. Explain admissibility and consistency of heuristic functions with suitable examples.
4. Describe important characteristics of greedy best-first search.
5. Illustrate with an example how priority queues and possible paths are tracked in greedy best-first search.
6. Describe the evaluation function used in A* search.
7. Elaborate the important decisions about A* search and also the problems with it.
8. Illustrate with an appropriate example the execution of the A* search algorithm.
9. Explain the distinguishing features of hill-climbing local search with an example.
10. How the search environment in a hill-climbing local search is described using ‘state space landscape’, and what measures are taken to face the odds of the landscape?
11. What ideas stand behind genetic algorithms?
12. Describe the major steps of a typical genetic algorithm and illustrate them with examples.
13. Narrate important features of ‘crossover’ and ‘mutation’.